

Name: Noman Tariq

Roll No:9702

Job Salary Prediction Using Random Forest

1. Project Objective

The project predicts the **expected salary** for a job based on features like experience, skills, and job role. It automates salary estimation and demonstrates the use of AI in real-world applications.

2. Dataset

- CSV file: salary_prediction_data.csv
 - Contains job-related features and corresponding salaries.
 - **Data Cleaning:** Removed missing values.
 - **Categorical Encoding:** Converted text features into numerical values using one-hot encoding.
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3. Features and Target

- **Features (X):** Job role, experience, skills, etc.
 - **Target (y):** Salary
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4. Model Used

- **Random Forest Regressor**
 - Ensemble of decision trees
 - Handles non-linear relationships
 - Reduces overfitting by averaging multiple trees
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5. Workflow

1. Load dataset and clean missing values.
 2. Encode categorical variables using one-hot encoding.
 3. Split data into **training set (80%)** and **testing set (20%)**.
 4. Train the Random Forest model on the training data.
 5. Predict salaries on the test set.
 6. Evaluate model performance using **R2 Score** and **Mean Squared Error**.
 7. Visualize feature correlations with a heatmap.
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6. Evaluation

- **R2 Score:** Indicates how well the model explains salary variance.
 - **Mean Squared Error (MSE):** Shows the average prediction error.
 - **Sample Prediction:** The model predicts salaries close to actual values.
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7. Visualization

- Heatmap shows correlation between features and salary.
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8. Conclusion

- The project successfully predicts salaries based on job-related features.
- Random Forest provides reliable predictions for non-linear, multi-feature data.
- Future improvements: Hyperparameter tuning, feature selection, and using larger datasets.